Unit 13, Lesson 3: Applying Properties of Operations

Benchmark

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

Learning Target

- ☐ I can apply the associative, commutative, and distributive properties to generate equivalent expressions.

13.3.1 Warm-Up: Mental Math

Ulrich finds the product of $5 \cdot 107$ in his head. Ulrich wrote the following explanation of his thinking:

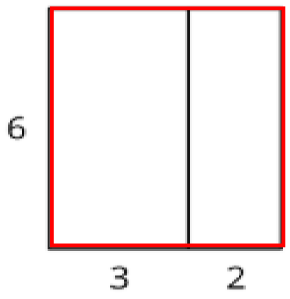
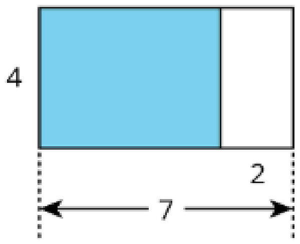
> Because 107 is equal to $100 + 7$, I can write $5 \cdot 107$ as $5(100 + 7)$. I know that $5 \cdot 100$ is 500 and that $5 \cdot 7$ is 35. Then, I add 500 and 35 to get 535.

1. Use Ulrich's reasoning to write each in the form $a(b + c)$. Then, use that form to find the product.

Multiplication	$5 \cdot 104$	$5 \cdot 97$	$5 \cdot 130$
$a(b + c)$			
Product			

13.3.2 Exploration: Representing Area of a Rectangle

1. Work with your partner to answer the questions.

Rectangle A	Rectangle B
	

a. Rectangle A is divided into two smaller rectangles. Select all the expressions that could be used to determine the area of Rectangle A.

- ☐ $6 \cdot 5$
- ☐ $6 \cdot 3 \cdot 2$
- ☐ $6(3 + 2)$
- ☐ $6 \cdot 3 + 2$
- ☐ $6 + 3 + 2$
- ☐ $6 \cdot 3 + 6 \cdot 2$

b. With your partner, discuss which expressions are in the form $a(b + c)$.

c. Rectangle B is divided into two smaller rectangles. Select all the expressions that could be used to determine the area of the blue rectangle inside Rectangle B.

- ☐ $4 \cdot 5$
- ☐ $4 \cdot 7 \cdot 2$
- ☐ $4(7 - 2)$
- ☐ $4 \cdot 7 - 4 \cdot 2$
- ☐ $4 \cdot 7 + 4 \cdot 2$
- ☐ $4 \cdot 2 - 4 \cdot 7$

d. With your partner, discuss which expressions are in the form $a(b + c)$.

2. Complete the following problems.

a. Circle all of the expressions that have the same value as $-4(x+7)$ if $x = {}^{-}13$.

$-4x+7$	$(x+7)\cdot -4$	$-4x+-4\cdot 7$	$4(x-7)$	${}^{-}28-4x$

b. With your partner, discuss whether or not the expressions you circled would change if the x-value were a different number. Summarize your decision and reason.

13.3.3 Guided Instruction: Applying Properties of Operations

The **distributive property of multiplication over addition** is a property that applies to all rational numbers. The distributive property "distributes" a factor to all terms within a set of parentheses.

1. Complete the table. The first row has been completed.

$a(b+c)$	$ab+ac$	Value
${}^{-}8(50+9)$	${}^{-}8(50)+{}^{-}8(9)$	${}^{-}472$
	$17(20)+17(3)$	
$(100-7)\cdot {}^{-}6$		
$(40+72)29$		
	${}^{-}11(48)-{}^{-}11(96)$	

2. In two of the rows the expression had a subtraction sign instead of an addition sign. Discuss with your partner how $a(b-c)$ can be rewritten using addition. Write a summary of your discussion.

3. Use the expressions below to show that the distributive property of multiplication over addition works with rational numbers.

$a(b+c)$	$ab+ac$	Equivalent?
$\frac{1}{2}(^{-}14+50)$	$\frac{1}{2}(^{-}14)+\frac{1}{2}(50)$	
$1.2(12+50)$	$1.2\cdot 12+1.2\cdot 50$	

Mathematicians determine a property exists when expressions are equivalent, no matter what number is used. A property can then be used with variables.

Expressions are **equivalent** if they have the same value, no matter what number is substituted in for each of the variables.

4. Write an equivalent expression for $8(x+2)$.

5. Write an equivalent expression for $\frac{1}{2}(x-16)$.

The **commutative property** states that the order in which an operation is performed does not change the value of the expression.

6. Use the pairs of expressions below to determine the operations for which the commutative property is true.

Operation	Example Expressions		Equivalent?
Addition	$\frac{5}{12}+\frac{3}{4}$	$\frac{3}{4}+\frac{5}{12}$	
Subtraction	$\frac{3}{4}-\frac{11}{2}$	$\frac{11}{2}-\frac{3}{4}$	
Multiplication	$\frac{5}{6}\cdot\frac{2}{3}$	$\frac{2}{3}\left(\frac{5}{6}\right)$	
Division	$\frac{7}{12}\div\frac{1}{4}$	$\frac{1}{4}\div\frac{7}{12}$	

The commutative property only applies to

__________________ and __________________________.

7. Use the commutative property of multiplication to write an expression equivalent to $5 \cdot \frac{3}{20} \cdot 4\frac{4}{5}$.

8. Rewrite $x + 9$ using the commutative property.

9. Rewrite $(^{-}7)(x)$ using the commutative property.

Commutative Property of Addition

$a + b = b + a$

Commutative Property of Multiplication

$a \times b = b \times a$

The **associative property** states that the way in which numbers are grouped does not change the value of the expression.

10. Use the pairs of expressions below to determine the operations for which the associative property is true.

Operation	Example Expressions		Equivalent?
Addition	$\left(\frac{8}{5} + \frac{3}{10}\right) + \frac{17}{10}$	$\frac{8}{5} + \left(\frac{3}{10} + \frac{17}{10}\right)$	
Subtraction	$\frac{7}{2} - \left(\frac{3}{4} - \frac{1}{4}\right)$	$\left(\frac{7}{2} - \frac{3}{4}\right) - \frac{1}{4}$	
Multiplication	$\frac{2}{3} \cdot \left(\frac{3}{5} \cdot \frac{1}{9}\right)$	$\left(\frac{2}{3} \cdot \frac{3}{5}\right) \cdot \frac{1}{9}$	
Division	$\left(\frac{1}{5} \div \frac{1}{10}\right) \div \frac{1}{2}$	$\frac{1}{5} \div \left(\frac{1}{10} \div \frac{1}{2}\right)$	

The associative property applies only to

______________________ and ___________________________.

11. Rewrite $(3.25 + 9.8) + 7.2$ using the associative property.

12. Rewrite $(x + 15) + {}^{-}11$ using the associative property.

13. Rewrite $({}^{-}2)(8 \cdot x)$ using the associative property.

14. Rewrite $6\big((x + 3) + 2\big)$ using the associative property.

Associative Property of Addition

$(a + b) + c = a + (b + c)$

Associative Property of Multiplication

$(a \times b) \times c = a \times (b \times c)$

13.3.4 Wrap-Up: Ticket Out the Door

1. Write an equivalent expression that uses each given property to complete the table.

Expression	Property	Equivalent Expression
$-2\big(4 + (x + {}^{-}7)\big)$	Commutative Property	
$-2\big(4 + (x + {}^{-}7)\big)$	Associative Property	
$-2(4 + x + {}^{-}7)$	Distributive Property	